

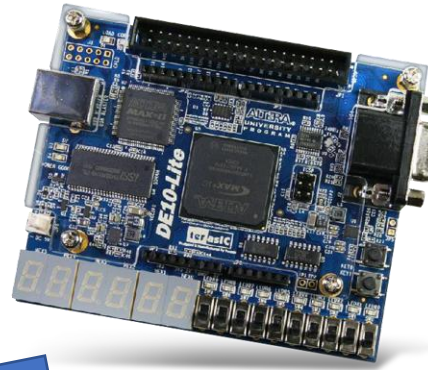
Final project – Home Automation

Unai Zurbano Martin - 873795621

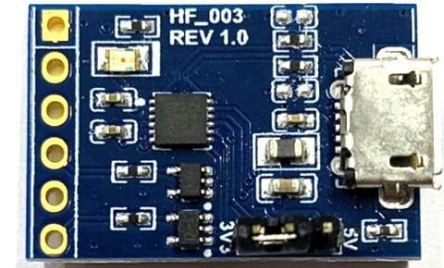
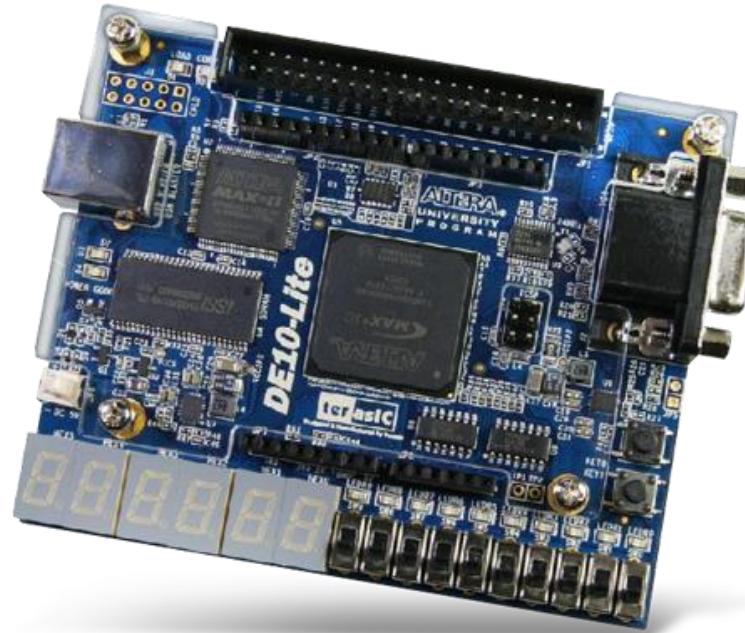
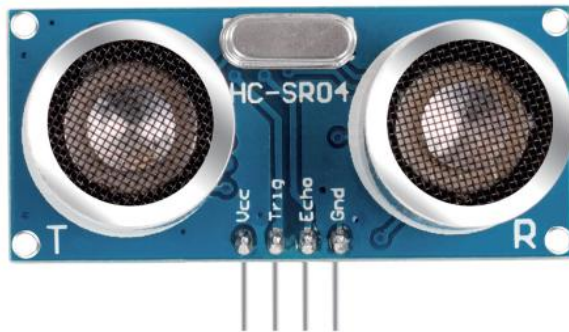
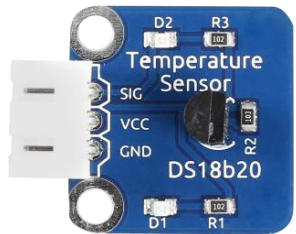
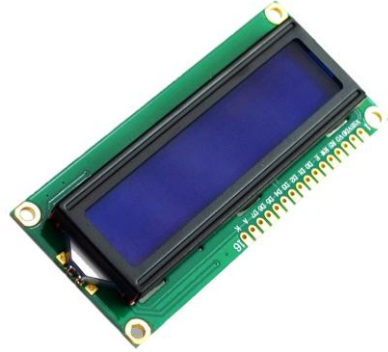
Concept



Concept



Components



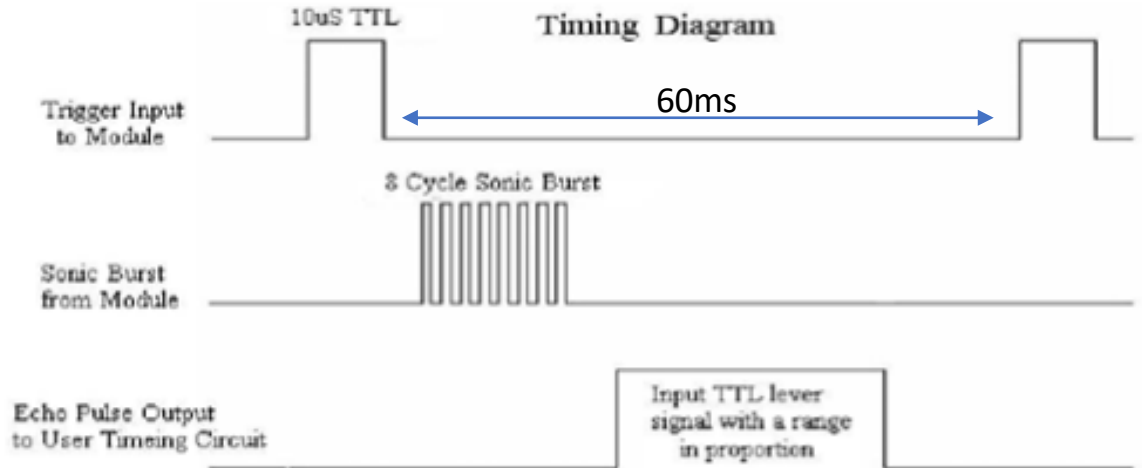
HC-SR04



Working Voltage	DC 5 V
Working Current	15mA
Working Frequency	40Hz
Max Range	4m
Min Range	2cm
MeasuringAngle	15 degree
Trigger Input Signal	10uS TTL pulse
Echo Output Signal	Input TTL lever signal and the range in proportion
Dimension	45*20*15mm

([datasheet](#))

- VCC: 5V
- Trigger: Input trigger
- Echo: Output measurement (Voltage divider)
- GND: Gnd



$$\mu\text{S} / 58 = \text{centimeters}$$

MQ-2



Potentiometer to adjust trigger digital output signal

- Detected Gas: Combustible Gas, Smoke
- Detection Intensity: 300 ~ 10000ppm(Combustible Gas)
- Working Voltage: 5.0V±0.1V AC or DC
- Preheating Time: ≥2 minutes

- D0: Digital output (Detection when 0) (Voltage divider)
- A0: Analog output Interface (AD module required)
- VCC: 5V (Voltage divider)
- GND: Gnd

([datasheet](#))

LC234X (UART interface)



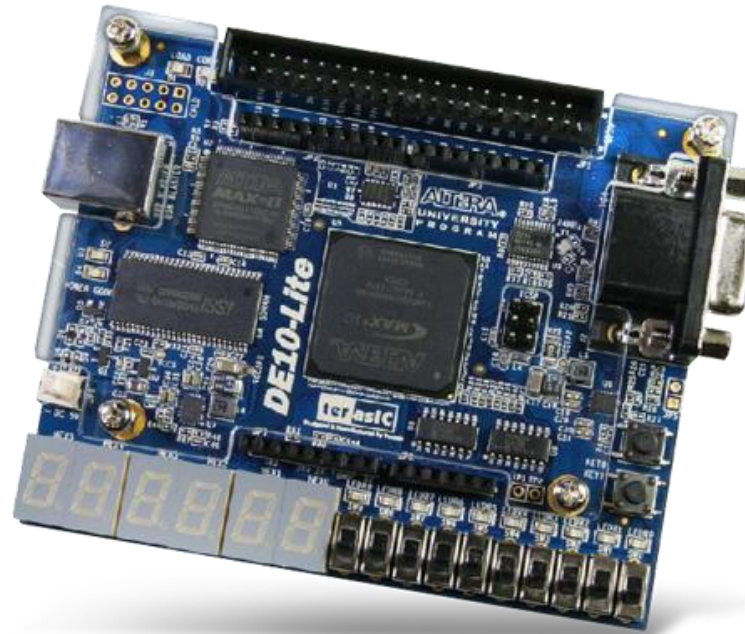
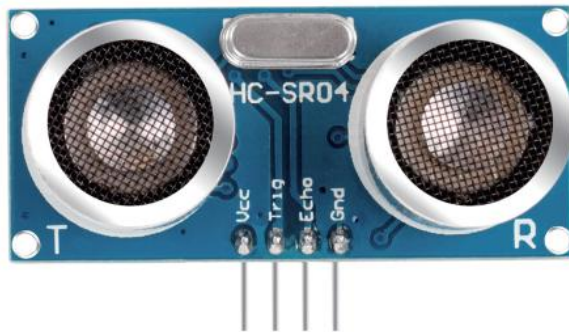
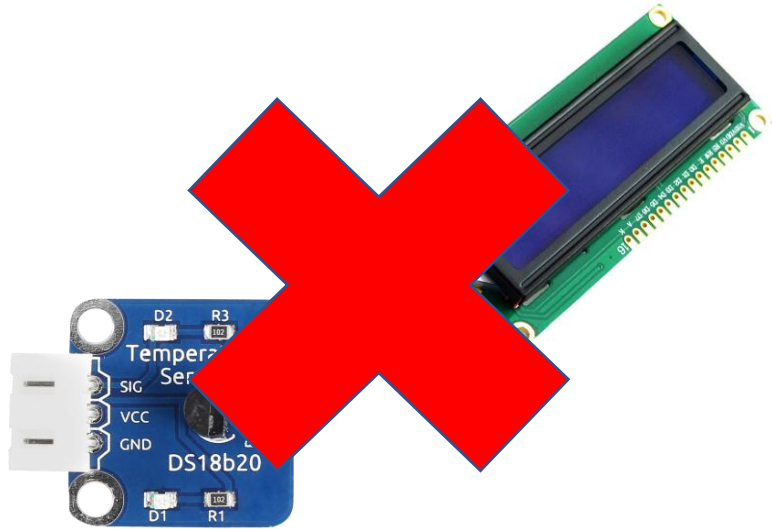
- Tx: Transmit Asynchronous Data Output
- Rx: Receiving Asynchronous Data Input
- GND: Gnd

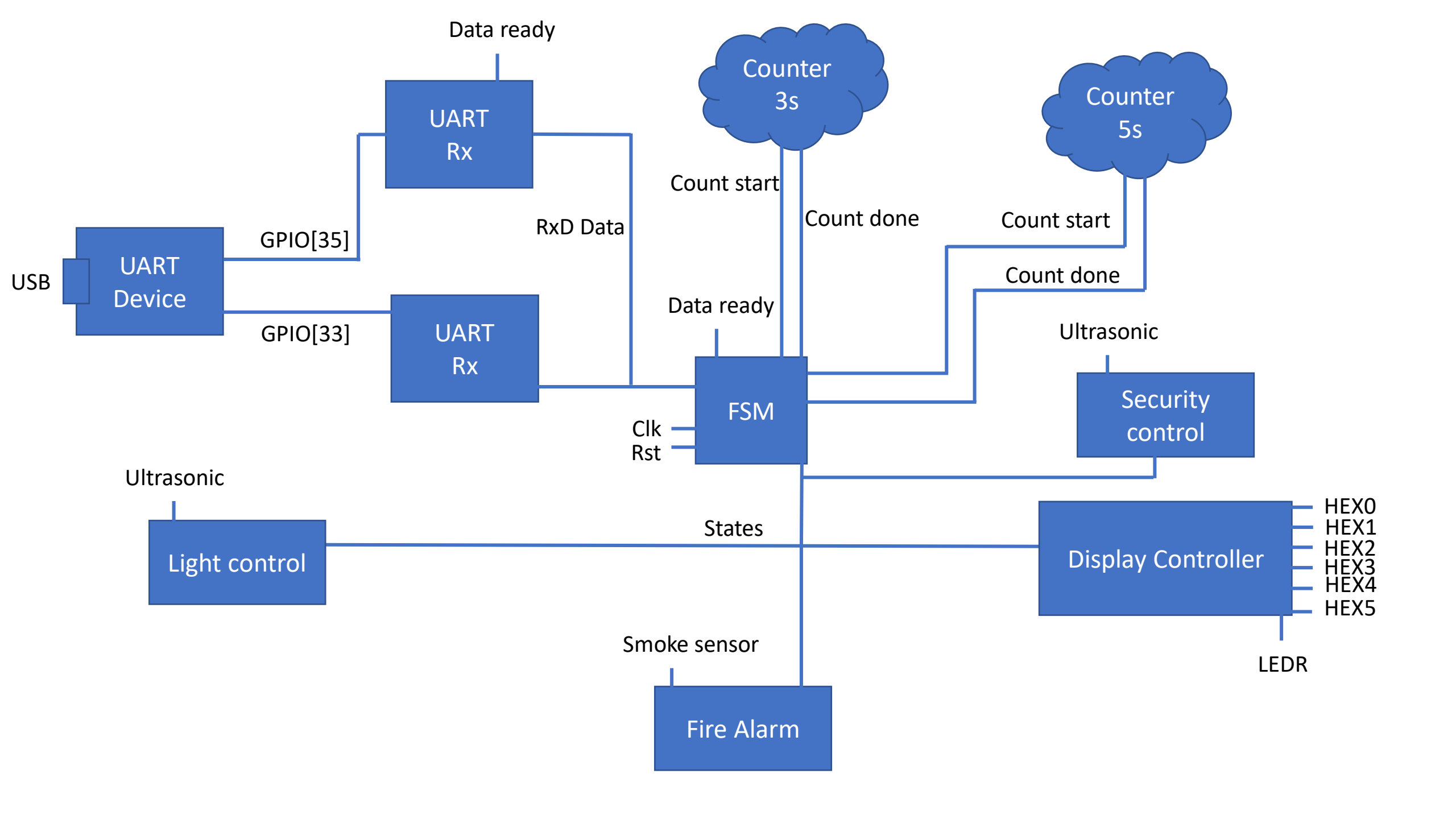
Feature	Reference Designator
Micro USB connector	CN1
IO voltage selection	CN2
6pin 2.54mm pitch connector UART interface	CN3
FT234X USB to serial bridge IC	U1
Green color LED	TX/RX

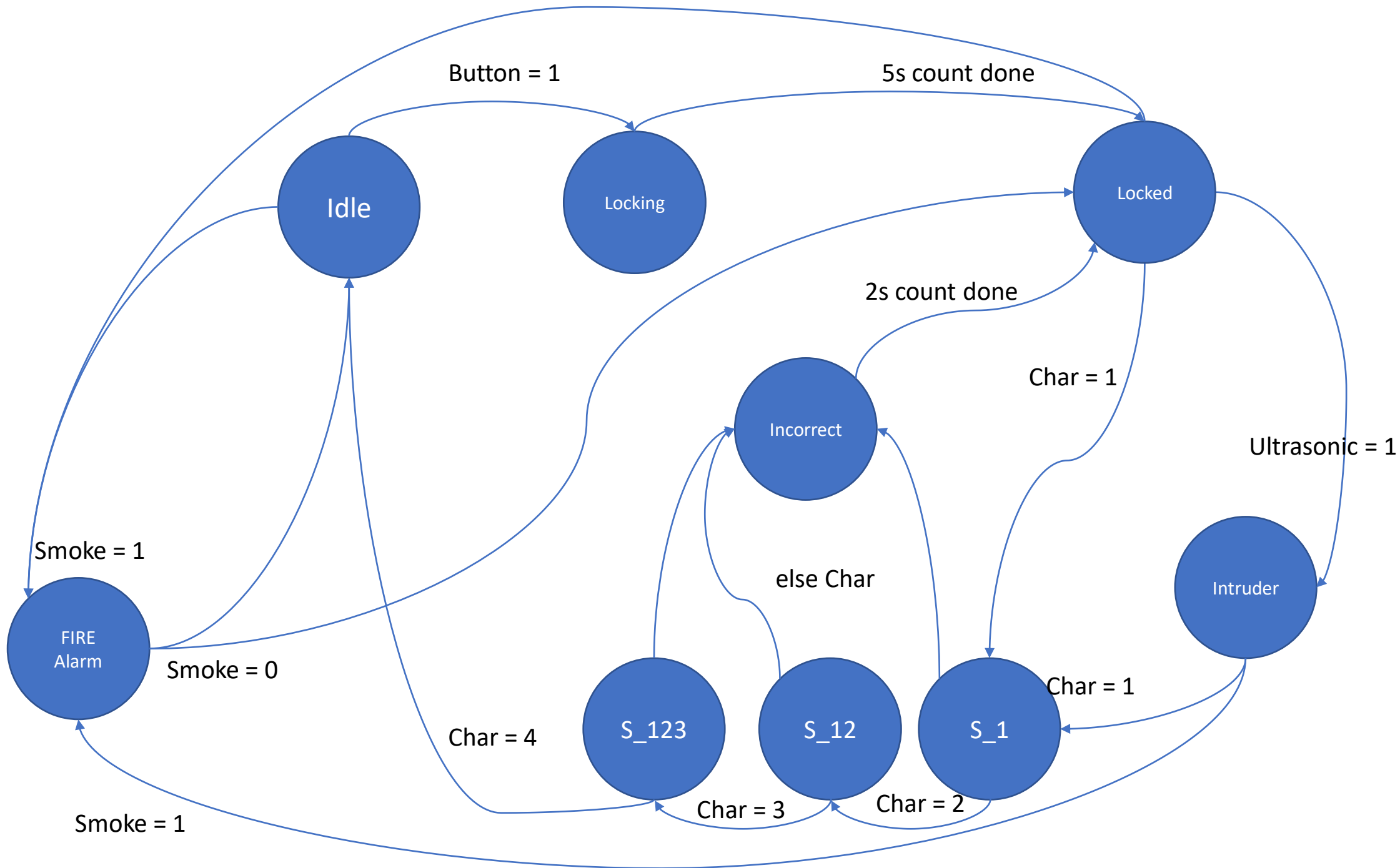
Table 1 - Module Features & Description

([datasheet](#))

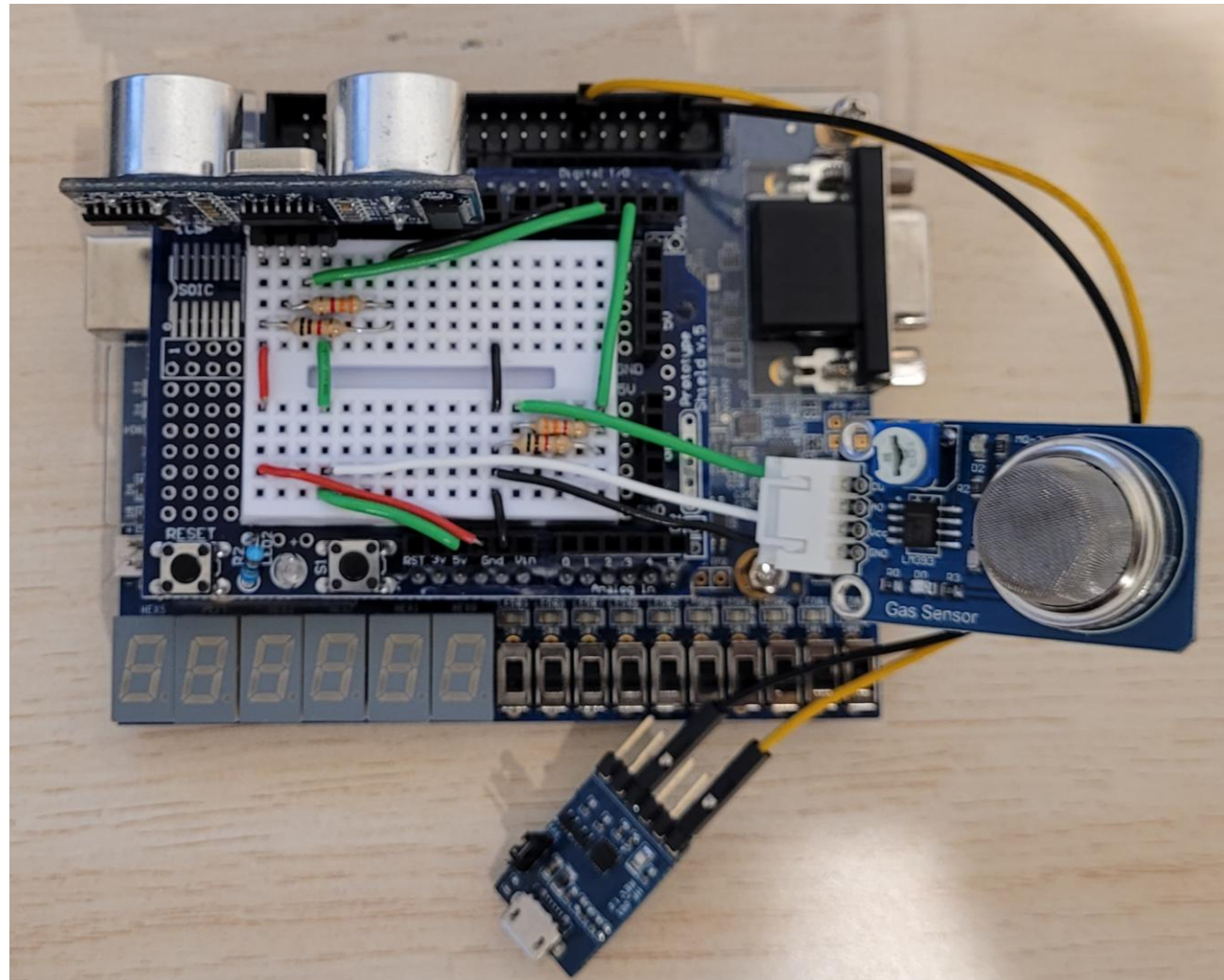
Components







Let's see it working!



Conclusion and future improvements

- Integration of multiple sensors. 
- FSM to manage security and real-time conditions 
- LCD Display and Temperature sensor 

Conclusion and future improvements

- Integration of multiple sensors. 
 - FSM to manage security and real-time conditions 
 - LCD Display and Temperature sensor 
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- Integration of remaining sensors
 - Add ADC for analog sensors
 - Possible integration with Bluetooth or Wi-Fi
 - 3D printed housing to reduce undesired conditions